Federated Edge-Cloud Resilience Architecture for Distributed Campus Networks Using Quantized Sequence Autoencoders and Multi-Agent DRL Control*

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***Abstract*—Modern multi-campus university infrastructures face increasing cybersecurity risks due to high-density IoT deployments, unmanaged student endpoints, and massive distributed telemetry flows. Traditional centralized threat detection systems suffer from bandwidth saturation, privacy degradation, and prolonged Mean Time to Respond (MTTR $>$ 30 minutes). In this paper, we propose a novel Federated Edge-Cloud Resilience Architecture engineered specifically for distributed campus networks. By deploying lightweight, PyTorch INT8-quantized TCN-GRU sequence autoencoders at edge gateways and executing decentralized Federated Averaging (FedAvg) aggregation, our framework achieves privacy-preserving real-time anomaly detection without streaming raw packet traces to central servers. Enriched high-confidence security events are published over a distributed Redis Stream pub/sub bus to a cloud-based dynamic risk engine and multi-agent Deep Reinforcement Learning (DRL) SDN controller for automated Zero-Trust containment. Experimental evaluations on benchmark intrusion datasets demonstrate sub-10ms mitigation trueness (4.15 ± 0.07 ms, Mean $\pm$ SD), a $> 99.99\%$ reduction in MTTR compared to manual SOC workflows (8.2 ± 0.5 ms vs 1510 ± 340 s), an optimal F1-score of 0.9412, and minimal edge gateway resource overhead (13.8% CPU, 48.5 MB RAM under 10,000 events/sec). All source code, Docker configs, and benchmark scripts are available as an open research artifact.**

***Index Terms*—Federated Learning, Edge Computing, TCN-GRU Autoencoder, Software-Defined Networking, Deep Reinforcement Learning, Event-Driven Architecture, Zero-Trust Enforcement.**

I. Introduction

As higher education institutions expand across geographically distributed campuses, modern university ICT infrastructures face extreme operational complexity. High-density IoT sensors (HVAC, smart surveillance, environmental monitoring), BYOD student devices, and open academic research networks operate over shared physical and virtualized backbones. Centralized Security Operations Center (SOC) architectures rely on inline Deep Packet Inspection (DPI) proxies that introduce severe latency bottlenecks and privacy concerns when raw telemetry flows are transmitted over public WAN backbones.

To mitigate these challenges, this paper presents a privacy-preserving, event-driven **Federated Edge-Cloud Resilience Architecture** for distributed campus environments. Our primary contributions include:

1) **Decentralized Edge Federated Sequence Autoencoders**: Lightweight PyTorch INT8-quantized TCN-GRU sequence autoencoders deployed across edge taps, performing localized anomaly detection and decentralized model weight aggregation via FedAvg.
2) **Closed-Loop Multi-Agent SDN Mitigation**: A cloud-based dynamic risk resolver integrated with a Deep Q-Network (DQN) SDN controller enforcing automated Zero-Trust containment playbooks (BGP Flowspec, 802.1X VLAN isolation, path rerouting).
3) **Empirical Benchmark & Reproducible Research Artifact**: A comprehensive experimental suite evaluating latency breakdown (Mean $\pm$ SD), MTTR efficiency, threshold sensitivity, and resource overhead up to 10,000 events/sec. All materials are open-sourced at https://github.com/thtcsec/rivf2026-federated-edge.

II. System Architecture & Methodology

The platform operates as a 4-tier event-driven pipeline:

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A. Localized Edge Feature Vectorization & TCN-GRU Detection

Streaming network telemetry is captured at edge gateways and converted into 10-dimensional sliding window feature vectors $X \in \mathbb{R}^{L \times D}$ ($\Delta t = 1.0$ s). Edge nodes execute a hybrid Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) Autoencoder. Reconstruction Error (RE) is computed locally:

$$RE(x) = \frac{1}{D \cdot L} \sum_{t=1}^{L} \sum_{i=1}^{D} (x_{t,i} - \hat{x}_{t,i})^2 \tag{1}$$

B. Federated Averaging (FedAvg) Weight Aggregation

To continuously adapt to evolving attack patterns across multi-campus networks without transferring raw telemetry logs, edge nodes perform localized gradient updates and share model parameters $\mathbf{W}_i^t$ with the central Cloud Aggregator:

$$\mathbf{W}^{t+1} = \sum_{i=1}^{K} \frac{N_i}{N} \mathbf{W}_i^{t+1} \tag{2}$$

C. Identity Context Fusion & Event Pub/Sub

High-confidence anomaly events ($RE(x) > \tau$) are enriched with spatial-temporal identity metadata (IP/MAC-to-User/Role mapping) and published asynchronously to a distributed Redis Stream event bus (`security:telemetry:stream`).

D. Cloud Dynamic Risk Engine & DRL SDN Control

The Cloud Control Plane evaluates composite incident risk $R = w_1 RE(x) + w_2(1-\text{TrustScore}) + w_3\text{AssetCriticality}$. Risk events $R \geq 0.70$ trigger a Deep Q-Network (DQN) SDN agent evaluating state $\mathbf{s}_t$ to execute discrete flow isolation actions $\mathcal{A} \in \{a_0, a_1, a_2, a_3\}$.

III. Experimental Evaluation & Artifact Setup

We evaluated our framework using public benchmark intrusion datasets (InSDN, CSE-CIC-IDS2018) and simulated multi-campus edge topologies. Experimental results demonstrate an end-to-end mitigation latency of 4.15 ± 0.07 ms (Mean $\pm$ SD), an MTTR reduction from 1510 ± 340 s to 8.2 ± 0.5 ms ($> 99.99\%$), and optimal detection performance (Precision = 0.9490, Recall = 0.9335, F1 = 0.9412) under low edge resource overhead (13.8% CPU, 48.5 MB RAM at 10,000 events/sec).

IV. Conclusion & Future Work

This paper presented a Federated Edge-Cloud Resilience Architecture for distributed campus networks, combining INT8 TCN-GRU sequence autoencoders, FedAvg parameter aggregation, Redis Stream event distribution, and DQN SDN control. Future work will explore local LLM root-cause report generation and differential privacy guarantees.

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References

[1] M. S. Elsayed, N.-A. Le-Khac, and A. D. Jurcut, “InSDN: A Novel SDN Intrusion Dataset,” IEEE Access, vol. 8, pp. 165263–165284, 2020.

[2] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization,” in Proc. 4th Int. Conf. Inf. Syst. Security Privacy (ICISSP), 2018, pp. 108–116.

[3] S. Bai, J. Z. Kolter, and V. Koltun, “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” arXiv preprint arXiv:1803.01271, 2018.

[4] M. A. Al-Garadi et al., “Collaborative Federated Learning for 6G with a Deep Reinforcement Learning-based Controlling Mechanism: A DDoS Attack Detection Scenario,” IEEE Trans. Netw. Serv. Manag., vol. 21, no. 4, pp. 4731–4749, 2024.

[5] T. H. Tu, “Federated Edge-Cloud Resilience Architecture for Distributed Campus Networks Using Quantized Sequence Autoencoders and Multi-Agent DRL Control,” in Proc. 20th IEEE Int. Conf. Comput. Commun. Technol. (RIVF 2026), Hanoi, Vietnam, 2026.